

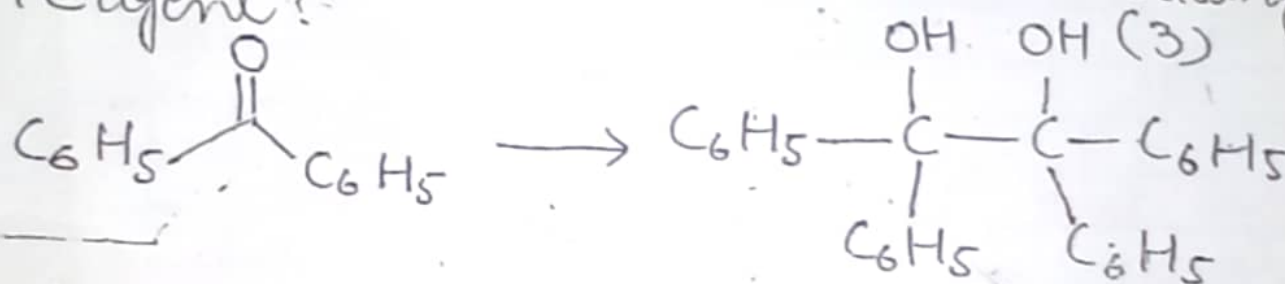
2. What is the final Product if cyclohexene reacts with KMnO_4 and then NaIO_4 ? (2)

23 (a) Define autoxidation and give its two industrial applications. (5) (255)

(b) Why ethers are explosives when used in Laboratory? (3) solve (203)

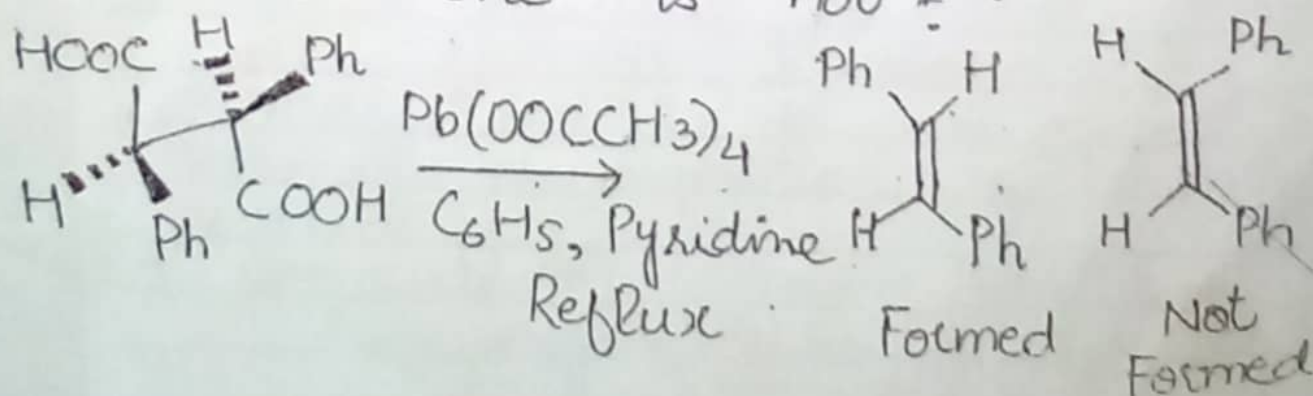
(c) Define hydrogenolysis with at least one example. (2) (269) (212)

4. (a) Complete the following reaction with mechanism and also write the missing reagent? (3)



b) Aniline absorbs at 232 nm. But in acidic solution the main absorption band is seen at 204 nm which is similar to benzene. Justify. (3)

c) Explain with mechanism that why one of the following compounds is formed and the other one is not?



(4)

Date _____

- Q 5. (a) Describe Oxidation of an aldehyde by
- (i) Jones's Reagent 25° (4)
 - (ii) Baeyer-Villiger ^{give}
- (b) Explain the Dakin's Reaction (3)
- (c) Describe the acid and base hydrolysis of the epoxides? (3)

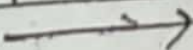
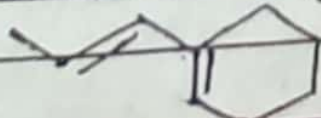
Six Semester 2014 Q No 1(a)

PAPER: Organic Chemistry 2(b)
3b(i)

Q 2: Short Answers:-

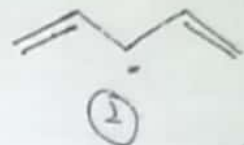
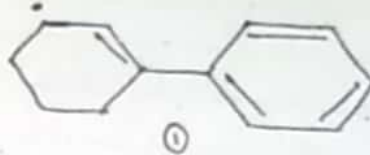
- I. Give Suitable reason that Why Cis-1,2-dichloroethene is infrared active while trans-1,2-dichloroethene is infra red inactive?
- II. What is mean by Finger Print region?
- III. The Energy difference between the ground state and excited State of an atom is 4.4×10^{-19} J. Calculate the λ required to produce this transition.
- IV. Define the term "persistent". What are radical Scavengers?
- V. What Product do you expect from Thermal decomposition of benzoyl peroxide?
- VI. Complete the following Reaction.

Perbenzoic acid ?



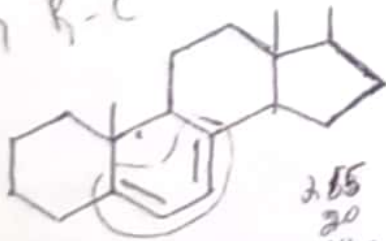
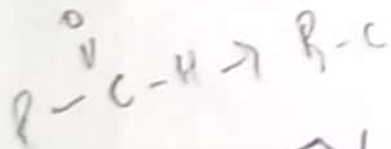
SMITHS

When 10ml of CCl_4 is added to 2ml of $\text{C}_2\text{H}_5\text{OH}$ then O-H (Str) is changed. Why? Arrange the following radicals in Order of increasing Stability. F.R. (4)

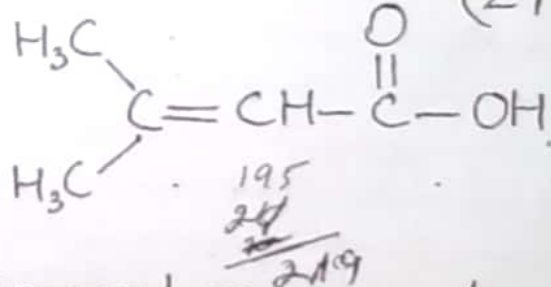


1. What is difference between Wolff-Kishner reduction and Clemmensen reduction?
 - Write a reaction for the conversion of o-toluidine (2-methylaniline) into o-aminobenzoic acid (anthranilic acid)?
 - Explain the following questions. (30)

Q.No.1 a) Predict the λ_{max} of the following Comp. (2+2)



2.15
20
10
24



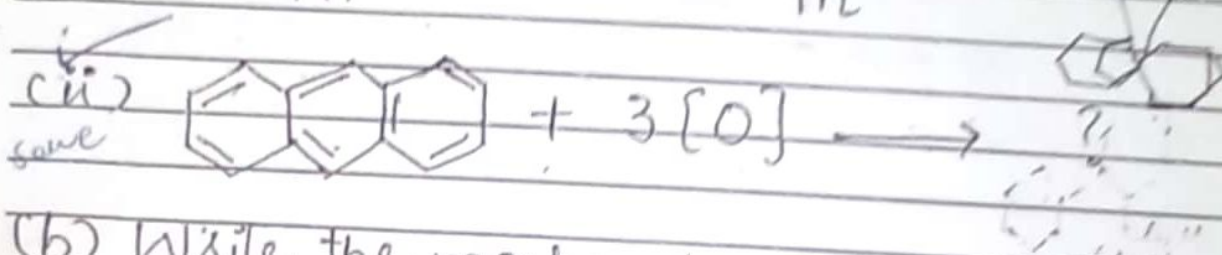
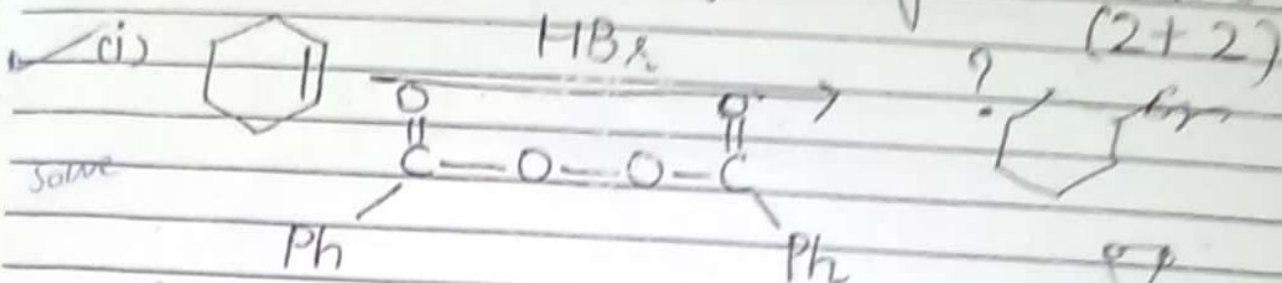
195
24
219

- (b) Define the terms chromophore and auxochrome, and explain with suitable examples. (4)
 c) What kinds of Vibrations are present in molecules. (2)

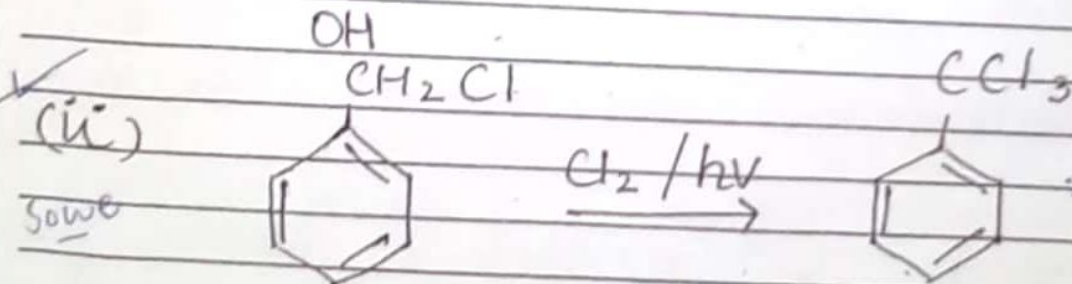
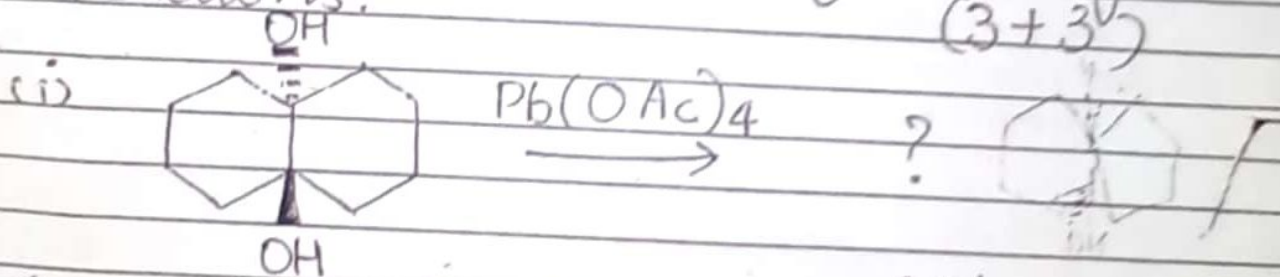
- Q.No.2 (a) Describe pinacolization of Ketones. (2)
 - (b) Write any two free radical general methods and briefly describe anti Oxidant applications of free radicals.
 c) Usually acidic and basic hydrolysis of the epoxides produce diols, Why? Repeat

Date _____

3. (a) Complete the following reactions



(b) Write the mechanism of following reactions.



SM 1000

UNIVERSITY OF THE PUNJAB

Sixth Semester 2015

Examination: B.S. 4 Years Programme

Roll No.

PAPER: Organic Chemistry
Course Code: CHEM-317

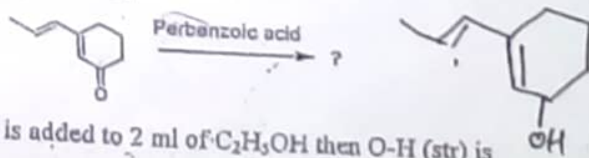
TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

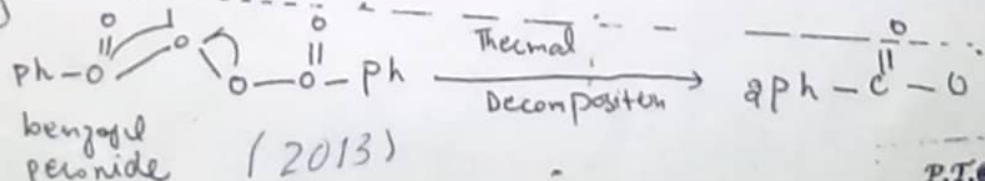
Subjective Marks: 50

Q.2 Answer the following questions briefly, all questions carry equal marks. (10 x 2 = 20)

- I. What is Baeyer-Villiger Oxidation? $260 \text{ R} \rightarrow \text{R}' + \text{CO}_2$
- II. Describe Mozingo method of reduction for ketones. $280 \text{ R} \rightarrow \text{R}' + \text{H}_2$
- III. What is Captodative Effect?
- IV. What is mean by finger print region? 1300 cm^{-1} (8-15.4)
- V. What is the difference between Wolff-Kishner reduction and Clemmensen reduction? *Net*
- VI. The energy difference between the ground state and excited state of an atom is $4.4 \times 10^{-19} \text{ J}$. Calculate the λ required to produce this transition. $\frac{1}{4.4 \times 10^{-19}}$
- VII. Define the term "persistent". What are *existing for long time* radical scavengers? *antioxidant e.g. vitamin*
- VIII. What product do you expect from thermal decomposition of benzoyl peroxide?
- IX. Complete the following reaction.



X. When 10 ml of CCl_4 is added to 2 ml of $\text{C}_2\text{H}_5\text{OH}$ then O-H (str) is changed. Why?



P.T.O.

UNIVERSITY OF THE PUNJAB

Sixth Semester - 2017

Examination: E.S. 4 Years Programme

Roll No. 012632

PAPER: Organic Chemistry
Course Code: CHEM-317

TIME ALLOWED: 2 hrs. & 30 min
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Q. NO. 2. Give the short answer of the following questions. ($4 \times 5 = 20$)

- I. Name different types of UV/Vis. transitions possible in Benzamide and Ethyl benzoate?
 - II. Why conjugation of C=O with C=C lowers its stretching frequency?
 - III. Describe Birch reduction with example and mechanism? 270
 - IV. Write a brief note on Wolf-Kishner reduction with example and mechanism. 278
 - V. Describe Oppenauer oxidation of secondary alcohols. Give example and mechanism. 254
- $R-CH_2-CH + Al(O-i-Pr)_3$

Q. NO. 3. Describe the following reactions with suitable examples and mechanism. (10)

- (i) Ozonolysis of Alkenes 236
- (ii) Hydroboration of Alkenes

Q. NO. 4. Discuss in detail the various types of vibrations and factors influencing these vibrations in IR Spectroscopy. (66 + 13) (10)

Q. NO. 5. Write a detailed note on the followings? (10)

- (i) Applications of UV/Vis. Spectroscopy in Chemistry
- (ii) Factors affecting stability of free radicals Solve

(2016)

cyclohexanone

(1) Na in liquid NH_3 , EtOH (2) m-chloro benzoic acid



UNIVERSITY OF THE PUNJAB

Sixth Semester - 2018

Examination: B.S. 4 Years Programme

Roll No. 15394

PAPER: Organic Chemistry

Course Code: CHEM-317 Part - II

TIME ALLOWED: 2 Hrs. & 45 M

MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Q. No. 2. Give the short answers of the following questions. ($4 \times 5 = 20$)

- I. What is Beers-Lambert law?
- II. What is the effect of ring size on IR absorptions in cyclic ketones?
- III. Describe Clemmensen reduction with example and mechanism? 276
- IV. Describe epoxidation of alkenes with example and mechanism? 227
- V. Describe two factors affecting the stability of free radicals?

Q. No. 3. Describe two different methods for the following conversions with mechanism?

- (i) Alkene to *Cis*-diol
- (ii) Acid chloride to aldehyde (5+5=10)

231-33

283

Q. No. 4. Write a note on the followings with example.

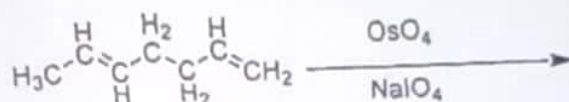
- a)- Infrared Spectrophotometer (Instrumentation) (5)
- ✓ b)- Hypochromic effect and Hyperchromic Effect (5)
Solve (2018)

Q. No. 5. Write a note on the followings?

- a)- Applications of UV/Vis. Spectroscopy? (5)
- ★ b)- Applications of free radical reactions? (5)

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED**Q.2. Answer the following questions.****(4x5=20)**

- Give possible electronic transitions in Ethyl benzoate and Acetaldehyde
- Discuss Birch reduction with suitable example and mechanism.
- Explain why ethylene does not show C=C stretching band while propylene does.
- What is role of free radicals in aging?
- Write product and mechanism of given reaction

**Q.3.**

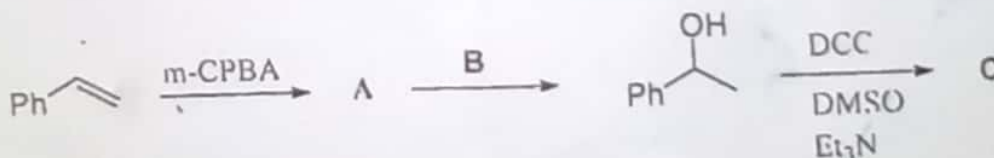
- What a note on Swern oxidation of alcohols? (5)
- Discuss different methods for oxidative cleavage of double bond? (5)

Q.4.

Describe sample handling in infrared spectroscopic technique? (10)

Q.5.

- How would you complete these reactions with mechanism? (6)



- Describe briefly hydroboration with mechanism and regeoselectivity? (4)

SHORT QUESTIONS (organic chemistry)

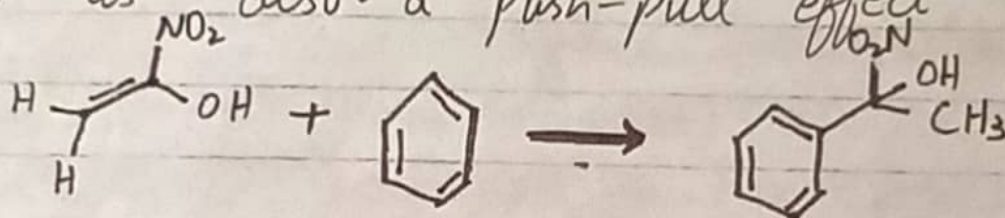
(2013)

(1)
What is captodative effect?

The captodative effect is the effect on the stability of a carbon-centred radical that is determined by the combined action of a captor (electron withdrawing) and a dative (electron donating) substituent.

→ both attached to the radical centre.

→ It is also called a push-pull effect



Captodative
olefin

Benze.

Friedel Craft
Product

(2)
Beer's law:

The law which describes the relationship b/w the absorbance of a solution and its concentration. It states:

"The amount of light absorbed is proportional to the concentration of the soln."

$$A = \epsilon LC$$

A = Absorbance

ϵ = molar absorptivity constant

L = Path length

C = Solution concentration (mol/L)

(3)

Describe bathochromic and hypsochromic shift:

1. Bathochromic shift:

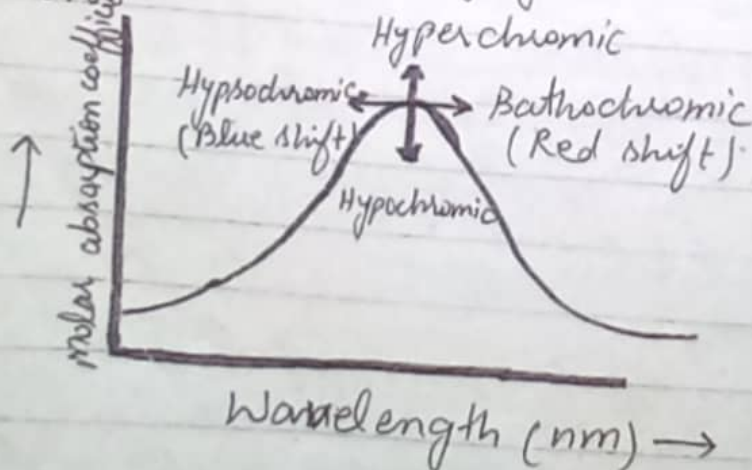
“Bathochromic shift is a change of spectral band position in the absorption, reflectance, transmittance, or emission spectrum of a molecule to a longer wavelength.”
→ As the red color in the visible spectrum has a longer wavelength than most other colors the effect is also commonly called **red shift**.

2. Hypsochromic shift:

“When ϵ of a compound is increased it is known as ϵ (When ϵ absorption intensity is increased it is known as ϵ)”

“When absorption maxima (λ_{max}) of a compound shifts to shorter wavelength it is known as hypsochromic shift or blue shift.”

→ This effect is due to presence of an group causes removal of conjugation or by the change of solvents.



(3)

(4)

Define term persistent? Which one radical will persist and why? Isopropyl / cyclohexyl

"the persistent radical effect (PRE) in chemistry describes and explain the selective product formation found in certain free-radical cross-reactions."

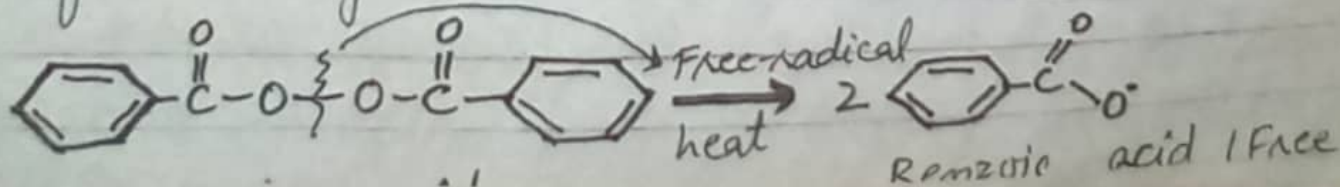
→ In these type of reactions, different radicals compete in secondary reactions.

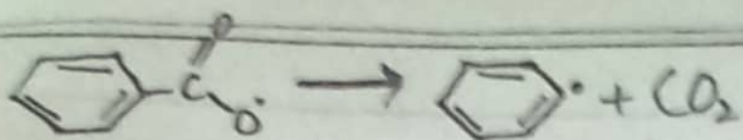
→ the so-called persistent (long-lived) radicals do not self terminate and only react in cross couplings.

(5)

What product do you expect from thermal decomposition of benzoyl peroxide?

The effect of temperature on the isothermal decomposition of benzoyl peroxide in solid medium shows the presence of other molecules decrease the activation energy of benzoyl peroxide decomposition of this peroxide in the presence of heat gives Benzene and Carbon dioxide.

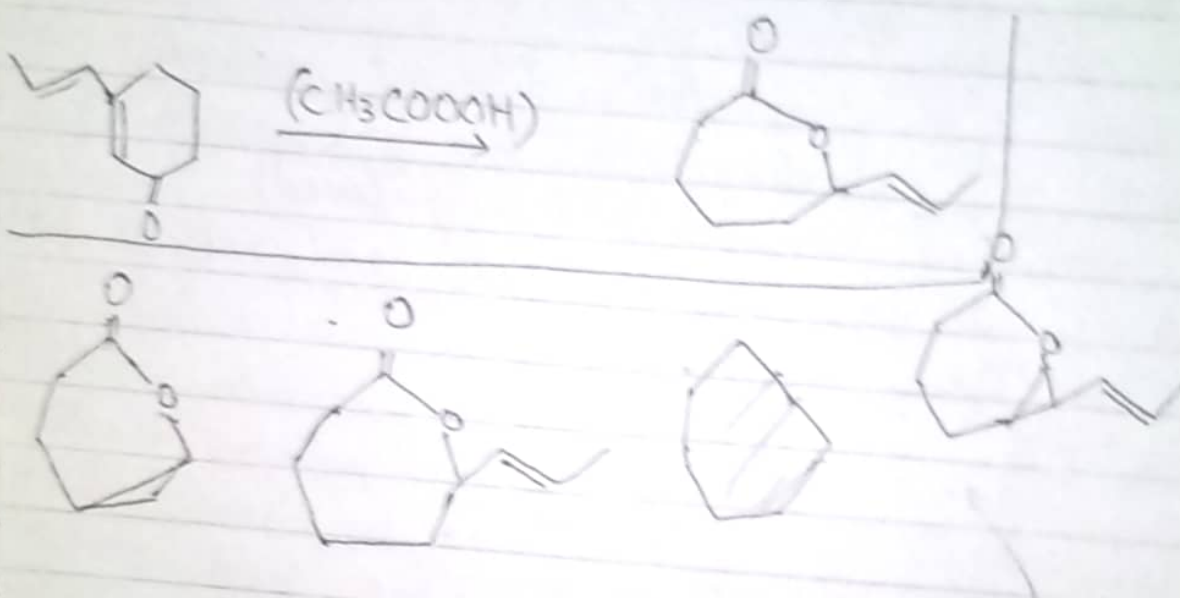
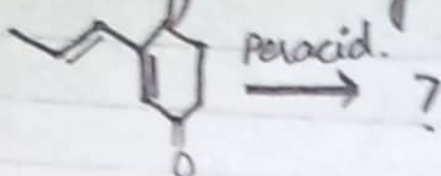




Benzoic acid

Benzene

(b)
Compare the following reaction.

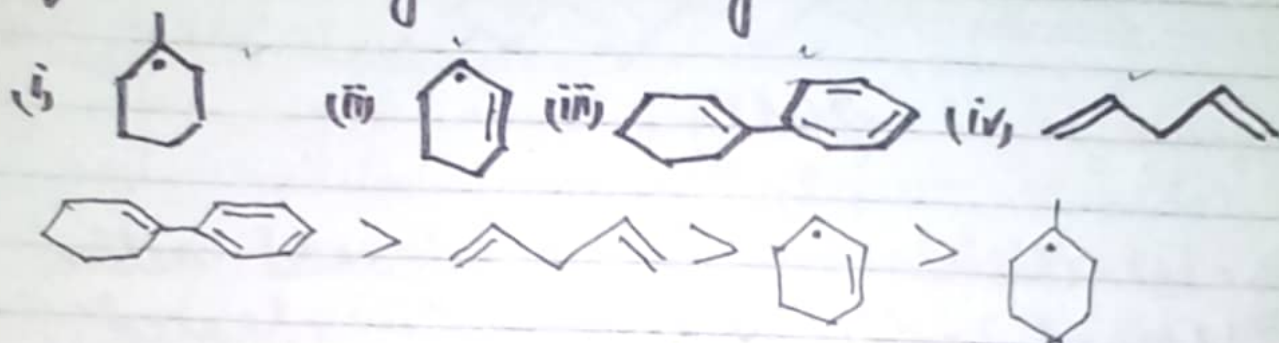


(7)
 NaBH_4 cannot reduce COOH group but boron reduce it to primary alcohol?
 COOH and esters are much less reactive to reduction than are ketones and aldehydes and sodium borohydride (NaBH_4) is too weak a reducing agent for them as compared to boron because acidic hydrogen is more reactive towards hydride ion than carbonyl carbon while on the other hand boron is strong reducing

(5)

agent that's why it reduces COOH to primary alcohol because the reactivity is too weak to occur without a stronger reducing agent.

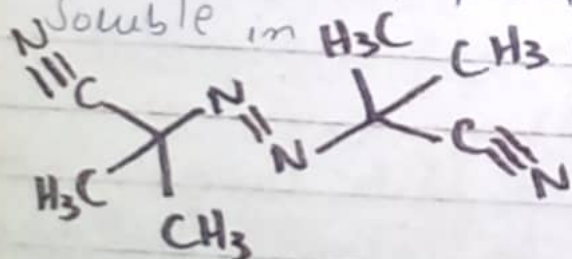
(8)
Arrange the following radicals in order of increasing stability.



(9)
AIBN and BHA abbreviated as?
AIBN BHA

It is abbreviated as Azobisisobutyronitrile

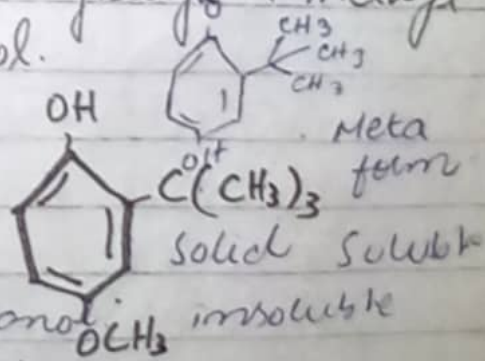
It is white powder
Soluble in alcohol



and common organic solvent
Soluble in water

It is abbreviated as 3-tertiary-butyl-4-hydroxy-anisole.

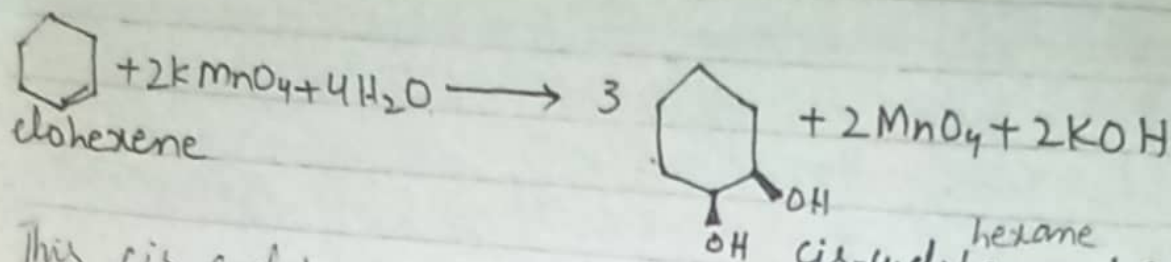
OR Butylhydroxyanisole
2-tertiarybutyl-4-methyl-phenol.



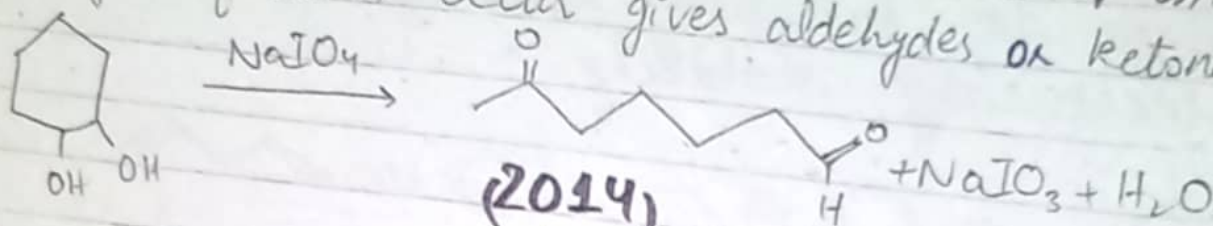
waxy in ethanol insoluble in water

(10)
What is the final product of cyclohexene reacts with KMnO_4 and then H^+ .

cyclohexene react with KMnO_4 cis-cyclohexene-1,2-diol formed.



This cis-cyclohexane-1,2-diol react with NaIO_4 and cleavage of diols occur gives aldehydes or ketones.

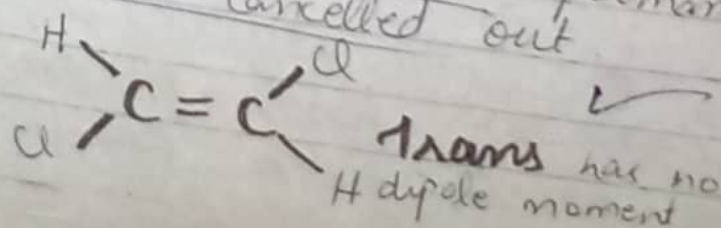
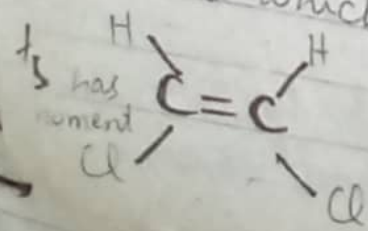


(2014)
(1)

My cis-1,2-dichloroethene is infrared active while trans-1,2-dichloroethene is infrared inactive:

An IR active bond is a bond that changes dipole during vibration while an IR inactive bond is a symmetric bond that doesn't change dipole during vibration. By the above information we can say that cis-1,2-dichloroethene is infrared active while trans-1,2-dichloroethene is infrared inactive because there will be no net change in the bond dipole moment.

any polarization in trans-product is cancelled out by a polarization in the opposite direction and the other hand cis-product has a permanent dipole moment which are not cancelled out.



(7)

(2)

What is meant by finger print region?

The region between 400cm^{-1} and 1500cm^{-1} in an IR spectrum is known as the finger print region.

→ It usually contains a large number of peaks, making it difficult to identify individual peaks.

→ In this region small difference in the structure and constitution of a molecule results in significant changes in the absorption bands.

Fingerprint region can be sub-divided into three types:

1. $1500-1350\text{cm}^{-1}$

It shows the presence of ter-butyl group in compound.

2. $1350-1000\text{cm}^{-1}$

Groups like alcohol, ester, lactones show absorption due to

3. Below 1000cm^{-1}

C-O stretching

Distinguishes b/w cis and trans alkenes.

(3)

The energy diff b/w ground state and excited state of an atom is $4.4 \times 10^{-19}\text{J}$.

Calculate λ required to produce this transition?

The energy diff b/w ground = $4.4 \times 10^{-19}\text{J}$
and excited state of atom

$$\Delta E = E_2 - E_1 = 4.4 \times 10^{-19}\text{J}$$

$$h = \text{Planck's constant} = 6.6 \times 10^{-34}\text{Js}$$

$c = \text{velocity of light} = 3.0 \times 10^8 \text{ m/s.}$
we know that.

$$\Delta E = \frac{hc}{\lambda}$$

$$4.4 \times 10^{-19} = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{\lambda}$$

$$\lambda = 4.5 \times 10^{-7} \text{ nm}$$

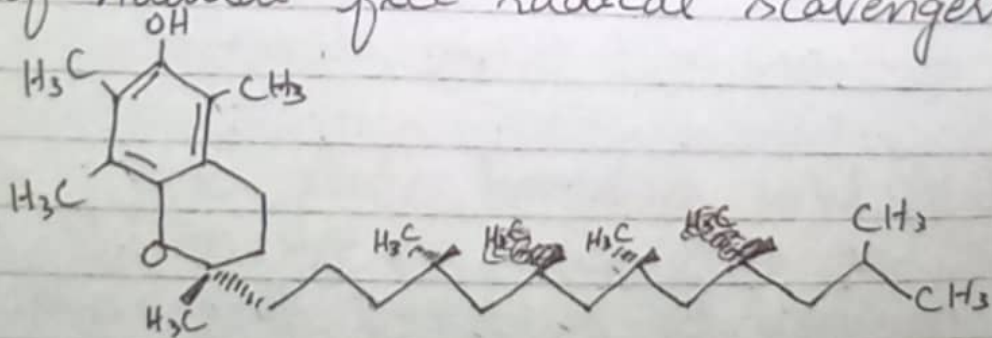
(4)
Define term persistent? What are radical scavengers:
→ Repeat.

Radical scavengers:

“These are molecules which can react with free radicals to form stable molecules and prevent chain reactions”

→ Free radical scavengers are added to food-products and to plastics.

Example: melatonin and Vitamin E are examples of natural free radical scavengers.



Vitamin E (α -tocopherol)

5,6. → Repeat

(7)
When 10ml of CCl_4 is added to 2ml $\text{C}_2\text{H}_5\text{OH}$ then O-H (str) is changed). Why?

Firstly CCl_4 and $\text{C}_2\text{H}_5\text{OH}$ are both are solvent. Secondly in $\text{C}_2\text{H}_5\text{OH}$ hydrogen bonding takes place. Hydrogen bonds are weak intermolecular. So OH group will be charged and replaced by chlorine atom.

(8) → Repeat
(9) ✓

Diff b/w Wolf-Kishner Reduction

→ An organic chemical reaction in which we convert a ketone or aldehyde into an alkane.

→ Occur in strong acidic conditions
Reduce carbonyl into

Clemmensen Reduction

→ An organic chemical reaction in which we convert carbonyl compound into methylene group.

→ Occur in strongly basic conditions.

→ Reduce carbonyl into

(4)
Specify important region in IR-spectrum

alkanes

methylene group

→ No catalyst is used.

→ Catalyst used is Zn/H₂O
(Amalgamated zinc)

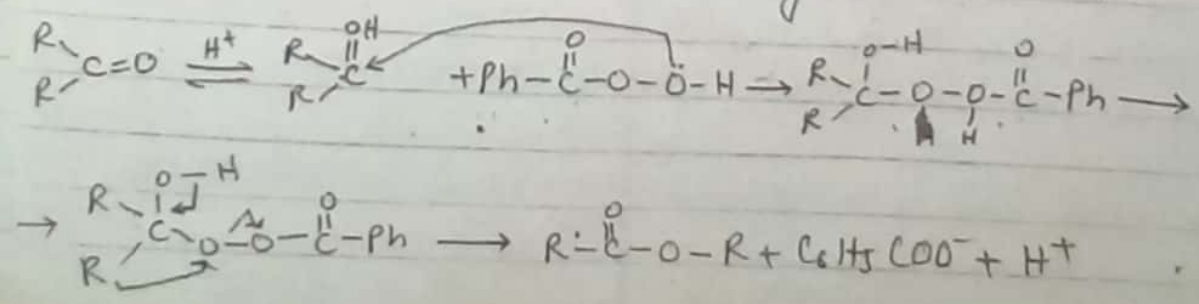
→ Unsuitable for ^{base} acid sensitive substrates

→ Unsuitable for acid sensitive substrate.

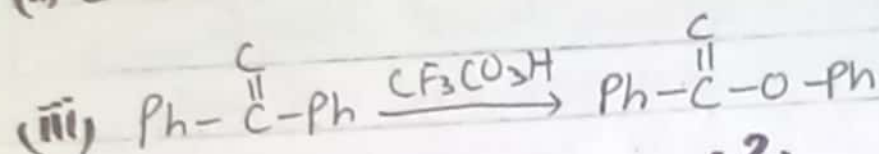
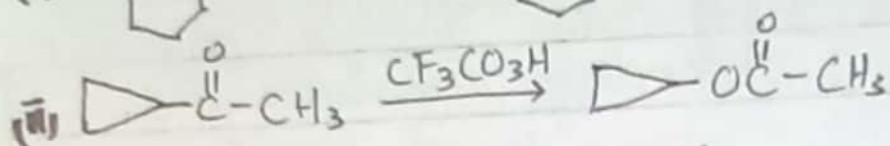
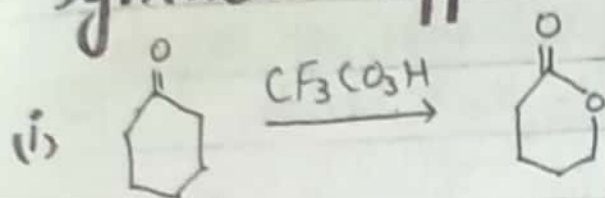
(10) Write a reaction for the conversion of o-toluidine (2-methylaniline) into o-aminobenzoic acid (anthranilic acid)?

E2 (2015)
Bayer-Villiger Oxidation:

Ketones are oxidized to ester and cyclic ketone into lactone under mild conditions with hydrogen peroxide and peracid. Perbenzoic acid, peracetic acid, m-chlorobenzoic acid, trichloroacetic acid are more commonly used.



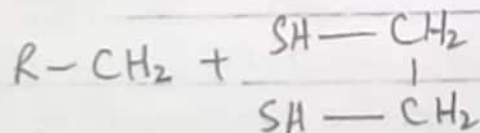
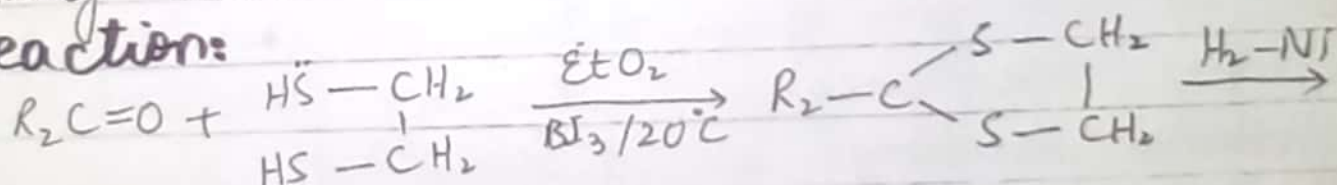
Synthetic Application:



(2) Meringo method of reduction for ketones:

In the 1st step of reduction the carbonyl compound is reduced with ethylene dithiol and Lewis acid or ketal with ethylene dithiol and Lewis acid and then this is further hydrogenolyzed with Raney nickel.

Reaction:



This method is useful for reducing acid or base sensitive carbonyl compound.

Remaining Questions of 2015 are repeat

3 → 10 (Repeat)

2016

(1) (All questions contains Four marks)

Diff b/w

Stretching vibration

Two or more bonds vibrates in and out together

Stretching Vibrations in which bond length changes that requires more energy

Distance b/w two atoms increases or decreases But the atoms are in same axis

There are two types of Stretching vibrations

1. Symmetric Stretching:

The movement of atom w.r.t particular atom is in the same direction.

2. Asymmetric Stretching:

In these vibrations one atom approaches the central atom while other departs from it.

There are two types of stretching vibration.

Bending Vibration

→ It changes the angle b/w two bonds.

→ Bending Vibrations in which bond length changes that requires less energy

→ The position of the atoms changes relative to the original bond.

Types:

→ There are Four types of Bending vibrations.

1. Scissoring:

Two atom approaches each other.

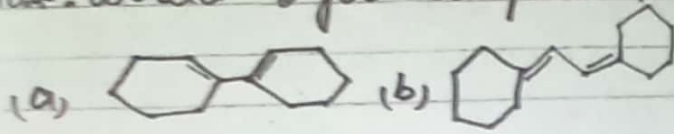
2- Rocking: The movement of atoms take place in same direction

3- Wagging: Two atoms move up and below the plane w.r.t the central atom

4- Twisting: One atom move up and other down the plane w.r.t central atom

(2)

How would you compare λ_{max} of following



λ_{max} refers to the wavelength along the absorption spectrum where a substance has its stronger photon absorption. Scientist use λ_{max} as a parameter to compare the different qualities of all types of molecules and substances.

Auxochrome

It is not whose absorption peak is shifted to the red end of the spectrum but it does shift the absorption peak to the red end of the spectrum.

The term chromophore is defined as the system which is responsible for imparting colour to the 2014 compound.

3(b) Chromophore:

①
"It is a Greek word (chroma = colour and phos = bearer). It is defined as any isolated covalently bonded group that shows a characteristic absorption of electromagnetic radiation in UV or visible region."

Compound containing chromophore is chromogen.

Example: $C=C$, $C=O$, NO_2

Types of Chromophore:

Independent

If one chromophore is required to impart colour

e.g.:

$-NO$, $-NO_2$, $-N=N$

Dependent

If more than one chromophore is required to impart colour.

e.g.:

Acetone, diketone.

Auxochrome:

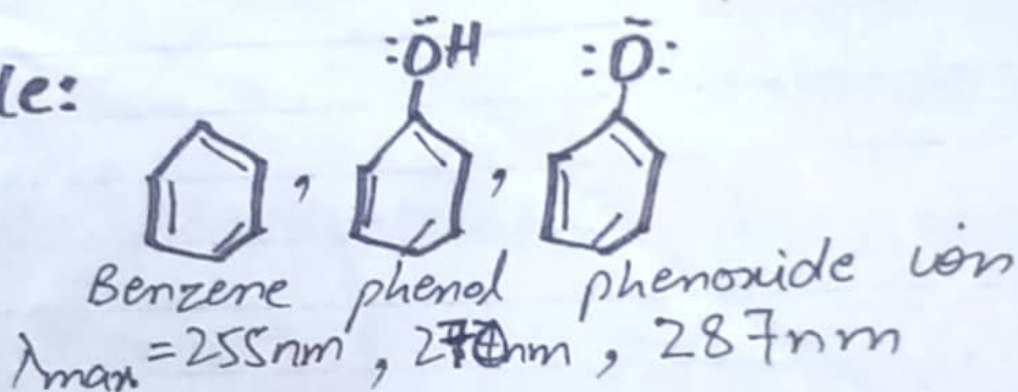
It is defined as any group which does not itself act as chromophore but whose presence brings about shift of the absorption band toward the red end of spectrum.

(2)

An auxochrome is a functional group of atoms with ~~no~~ nonbonded electron which when attached to a chromophore alters both the wavelength and intensity of absorption.

A Feature of these auxochromes is the extending of the conjugated system by resonance.

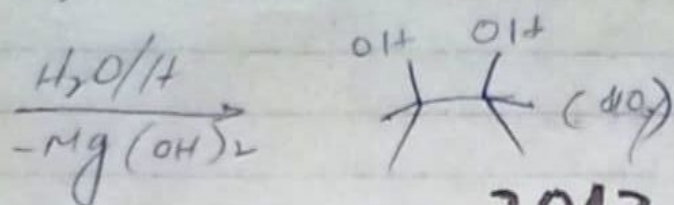
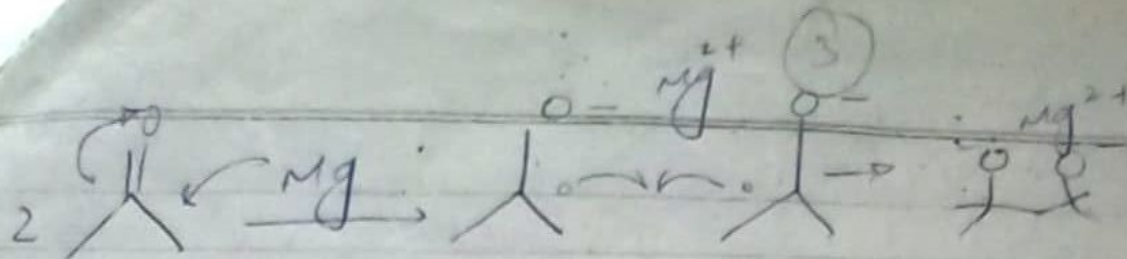
Example:



4-a) Pinacolization of ketones:

“A pinacol coupling reaction is an organic reaction in which a carbon-carbon covalent bond is formed b/w the carbonyl groups of an aldehyde or a ketone in presence of electron donor in a free radical process.”

The reaction product is vicinyl diol. The reaction is named after pinacol (also known as 2,3-dimethyl-2,3-butane diol or tetramethylethylene glycol) which is the product of this reaction when done with acetone as reagent.

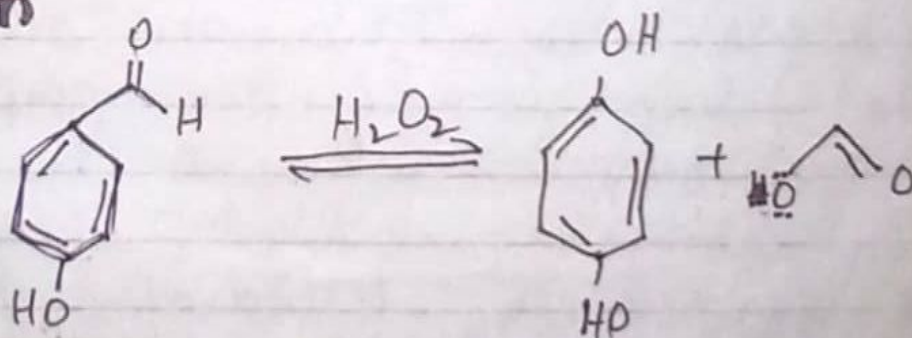


Dakin Oxidation: 2013 ✓

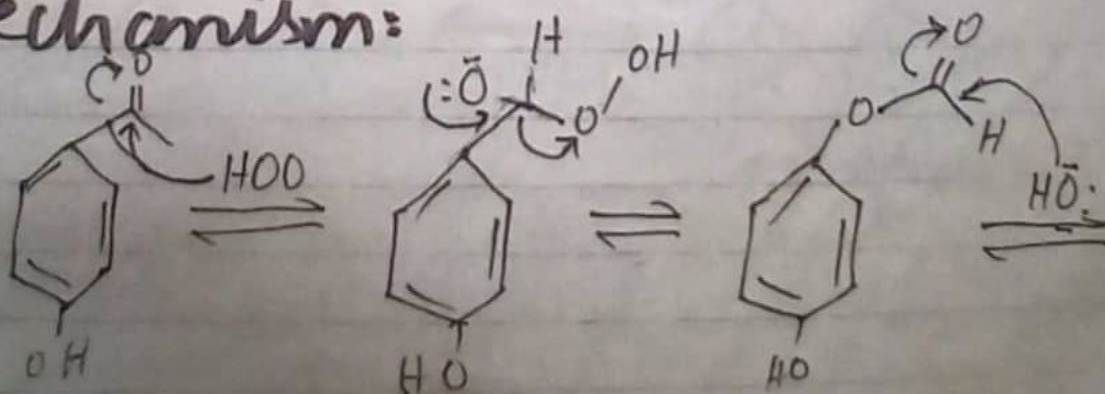
The Dakin oxidation is an organic redox reaction in which an ortho or para-hydroxylated phenyl aldehyde or ketones react with hydrogen peroxide in base to form a benzenediol and a carbonylate.

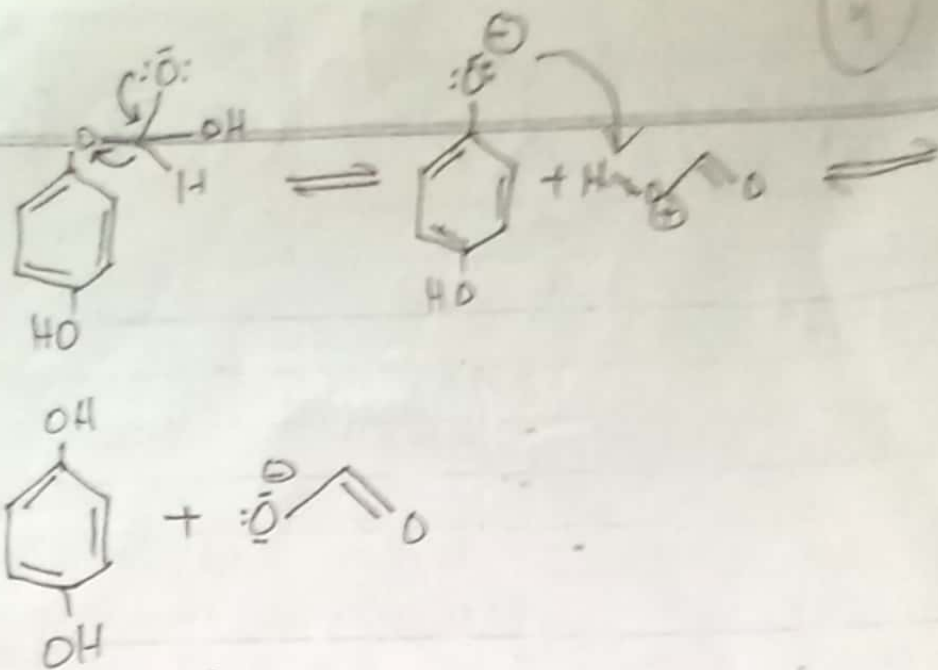
→ Carbonyl group is oxidized and H_2O_2 is reduced.

Reaction



Mechanism:



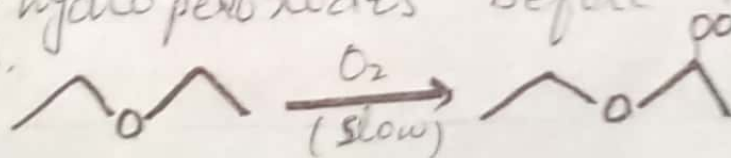


3(b) Why ethers are explosives when used in laboratory:

Ethers represent a class of material which can become more dangerous with prolonged storage because they tend to form explosive peroxides.

Exposure to light and air enhance the formation of the peroxides.

Hydroperoxides can be explosives so laboratory samples of ether must be frequently tested for the presence of hydroperoxides before they are used.

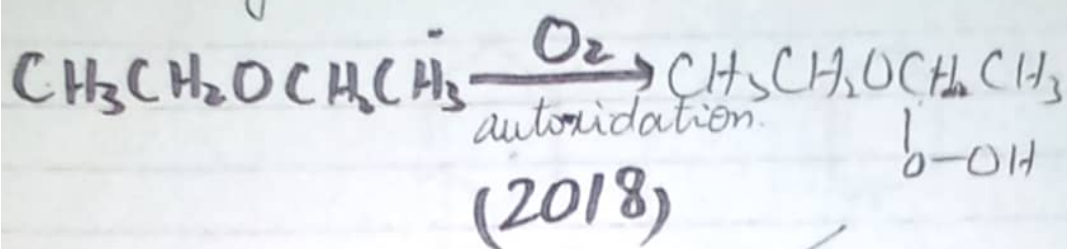


Example:

Diethyl ether is a very low boiling, highly flammable liquid. Care should always be taken when

(5)

diethyl ether is used in the laboratory because open flames or sparks from light switches can cause explosive combustion of mixtures of diethyl ether and air.



Hypochromic effect:

It is defined as an effect due to which the ~~coefficient~~ ϵ_{max} intensity of absorption maximum decreases i.e. extinction coefficient (ϵ_{max}).

Example: biphenyl absorbs at 250 mμ, $\epsilon_{\text{max}} = 19000$ whereas 2-methyl biphenyl absorbs at 237 mμ, $\epsilon_{\text{max}} = 10250$ (ϵ_{max} decreases). Introduction of methyl group distorts the geometry of the molecules, thus, cause hypochromic effect.

Hyperchromic effect:

It is an effect due to which the intensity of absorption maximum increases.

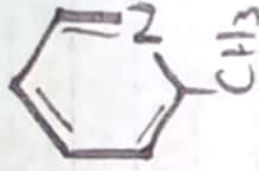
Example:

The B-band for pyridine

at 257 mμ, $\epsilon_{\text{max}} = 2750$ is shifted to 262 mμ, $\epsilon_{\text{max}} = 3560$ for 2-methylpyridine. Introduction of an auxochrome usually increases intensity of absorption.



pyridine



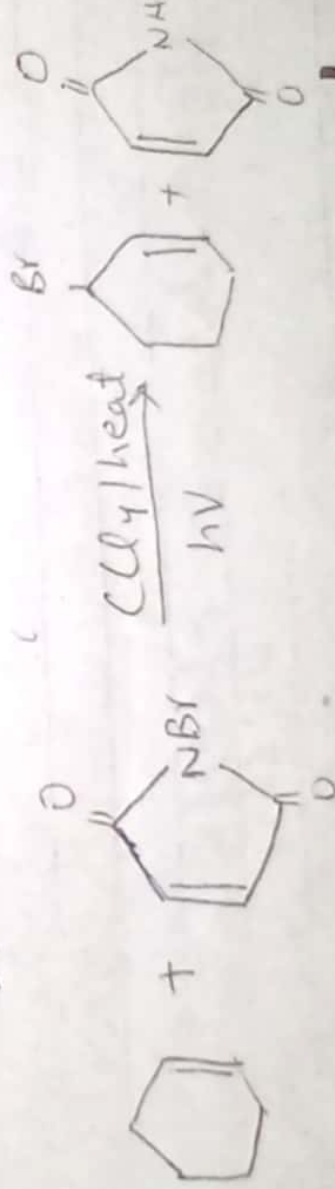
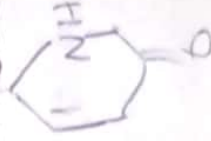
2-methyl pyridine

(2014)

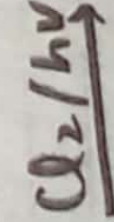
3(a) Complete the following Reactions:



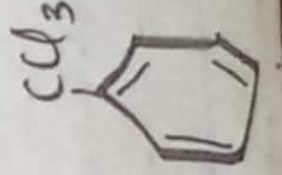
?



(b) Write the mechanism of following:



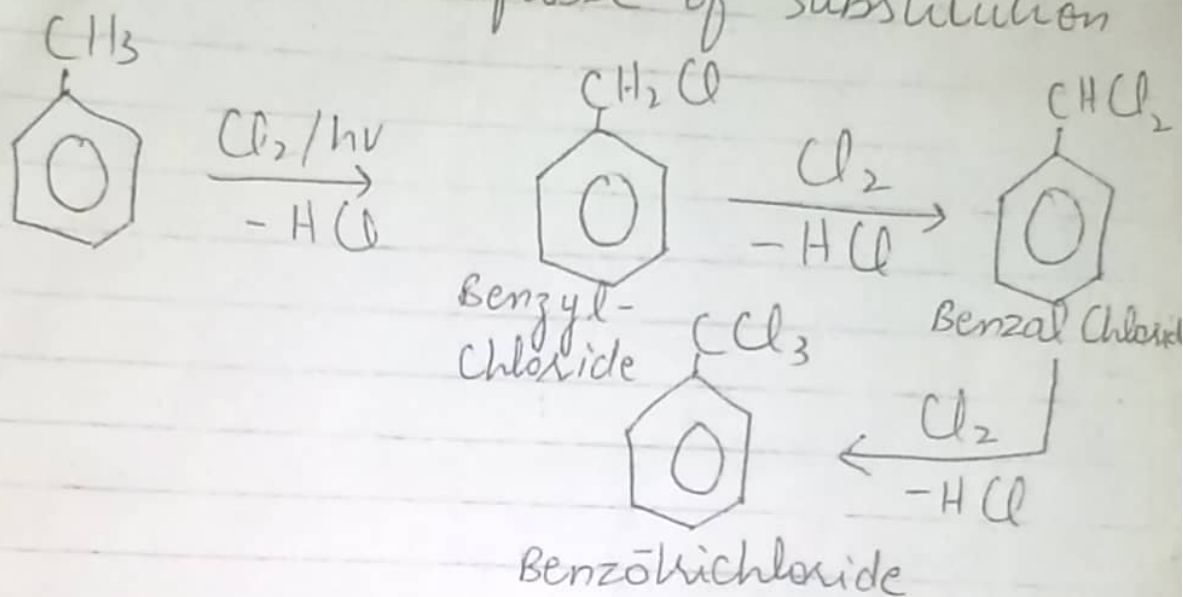
(ii)



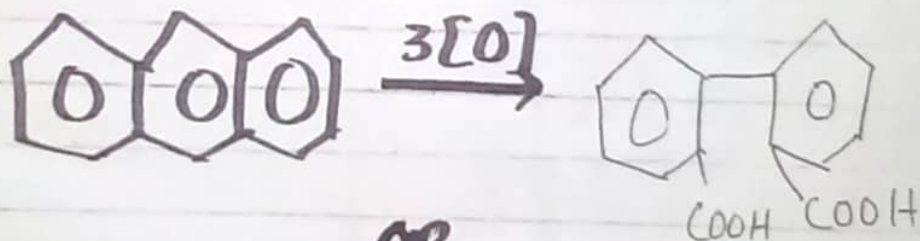
the mechanism of

Mechanism:

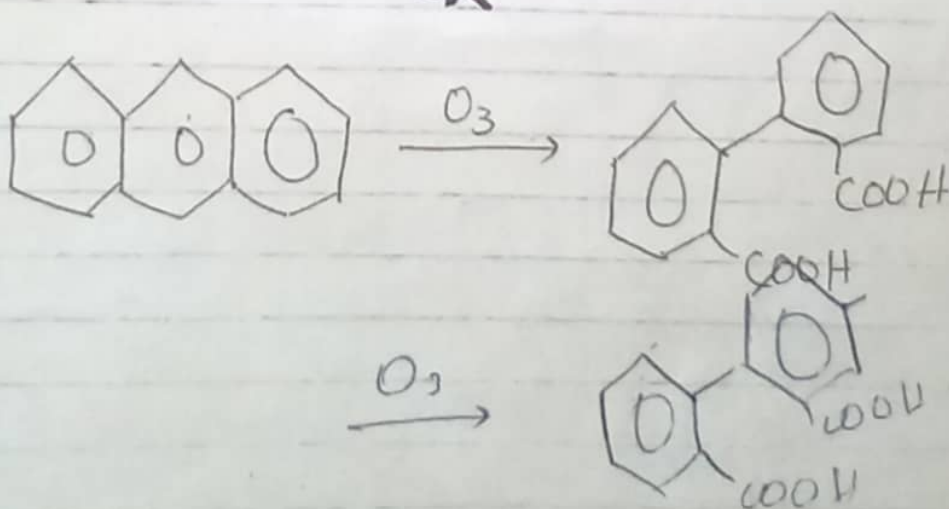
Radical halogenation can yield a mixture of halogenated compounds because all hydrogen atoms in alkanes are capable of substitution



3(a)ii)



OR

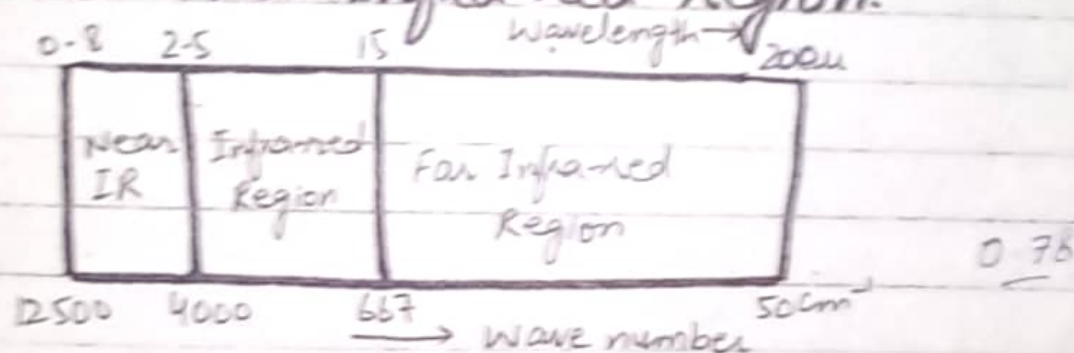


(4)

Specify important region in IR-Spectrum and explain their significance:

There are many region in IR-Spectrum. The most important region for an organic chemist is $2.5\mu - 15\mu$ in which molecular vibrations can be detected and measured in an Infra-red spectrum and in a Raman Spectrum.

The ordinary IR region extends from $2.5\mu - 15\mu$. The region from $0.8\mu - 2.5\mu$ is called Near Infra-red region. and then from $15\mu - 200\mu$ is called Far-Infra-red region.



(Infra-red Region)

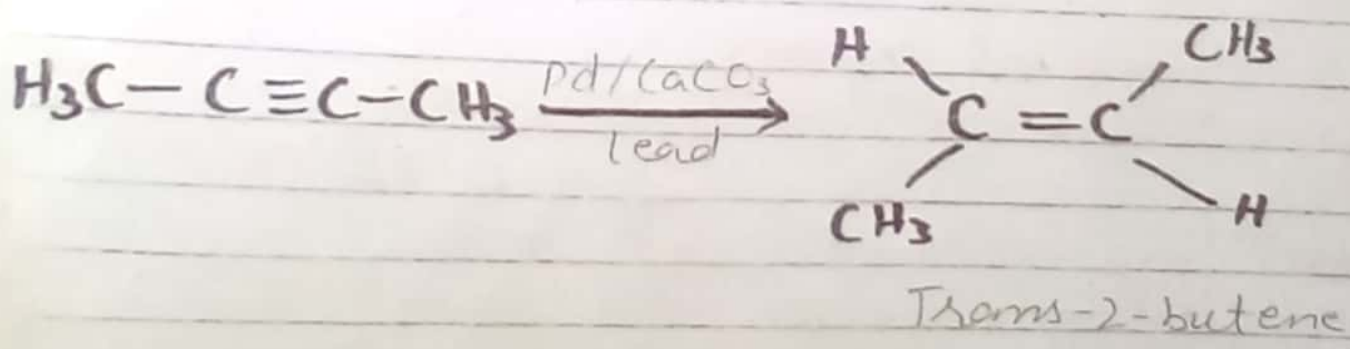
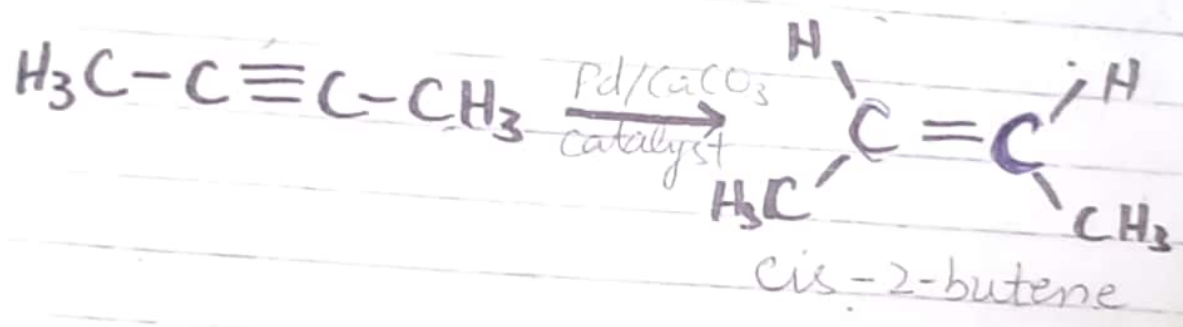
The term "Infra-red" covers the range of the electromagnetic spectrum between 0.78 and 1000mm . Infra-red spectroscopy, and wavelength is measured in wave numbers, which have the units cm^{-1} . The most useful IR region lies between $4000 - 670\text{cm}^{-1}$. It covers a range of techniques mostly based on absorption spectroscopy. With all spectroscopy technique it can be used to identify and study chemical substance. Samples may be solid, liquid

... vibrational energy will change which cause ~~transitions~~ ^{vibrations} ~~are~~ displayed in the form of bonds

called IR Spectral bands. In gases, solids and liquids only vibrational bonds

How can you reduce ~~any~~ ^{all} ~~the~~ 2-butyne into cis-butene and Trans-2-butene? ^{Structure of functional}

A Lindlar catalyst is a heterogeneous catalyst that consist of Palladium deposited on calcium carbonate which is then poisoned with various forms of lead or sulphur. It is used for hydrogenation of alkynes to alkenes (without further reduction into alkane) and is named after its inventor **Herbert Lindlar**.



(2017)

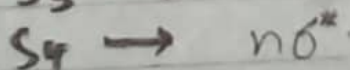
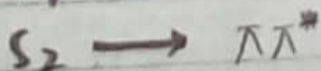
(1)

Name different types of UV/vis transition possible in Benzamide and Ethyl benzoate.

UV/vis spectra of ethyl benzoate and shows that ethyl benzoate had a four absorption peaks in the scanned range. The highest absorption peak was at **228nm** which was π -band peak generated by $\pi \rightarrow \pi^*$ transition of the large conjugated system of the whole molecule. The absorption peak at **205nm** was π -band peak of benzene ring. Additionally the absorption peak at **272** and **280nm** was π -band generated by $\pi \rightarrow \pi^*$ transition and vibration effect.

$S_0 \rightarrow S_2$ and $S_0 \rightarrow S_3$ transition are due to the MO of the π -type and are clearly pronounced $\pi \rightarrow \pi^*$ transitions. An intensive absorption of benzamide result from the allowed $S_0 \rightarrow S_3$ transition. The substitution by benzamide results in the shift into the long wavelength region. The absorption band intensity of halogens derivatives at **260-270nm** is due to $S_0 \rightarrow S_2$ transitions.

Characteristics of Electronic transition for free molecules of Benzamides.



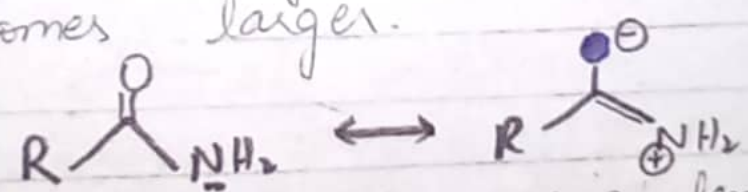
Why its conjugation of $C=O$ with $C=C$ lowers stretching frequency?

The carbonyl shows strong $C=O$ stretch absorptions between 1850 cm^{-1} and 1640 cm^{-1} . This is an IR band that can be detected.

The carbonyl stretch is sensitive towards its environment. A typical ketone or aldehyde carbonyl shows a strong $C=O$ stretch at around 1715 cm^{-1} . Electron withdrawing group lead to a $C=O$ stretch at larger wavenumbers, whereas electron-donating group leads to smaller wavenumbers.

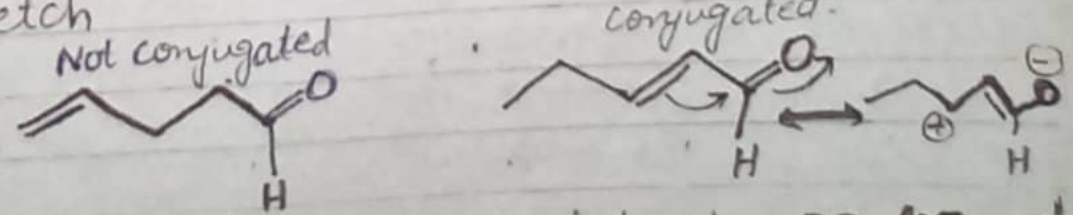


[Inductive effect] of an electronegative substituent shortens and strengthens $C=O$ bond. The $C=O$ frequency becomes larger.



The +M effect of an amino group lengthens and weakens the $C=O$ bond.

Conjugation with $C=C$ double bonds affects $C=O$ stretch

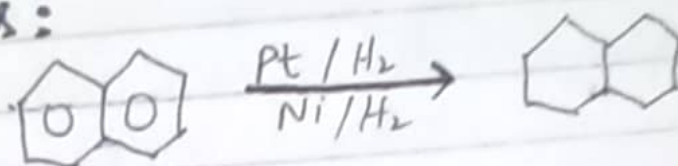


Conjugation lowers $C=O$ stretch by $20-40\text{ cm}^{-1}$ and lengthens and weakens the $C=O$ bond.

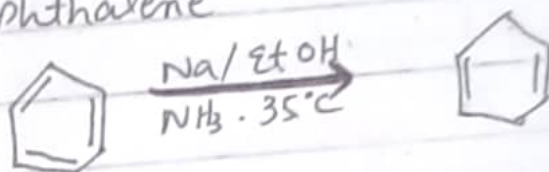
(3) Birch Reduction with example and mechanism

Birch Reduction is used to convert aromatic compound having benzoid ring into a 1,4-cyclohexadiene. This reaction is performed with metals like sodium, lithium and potassium in alcohols or liq ammonia.

Reactions:

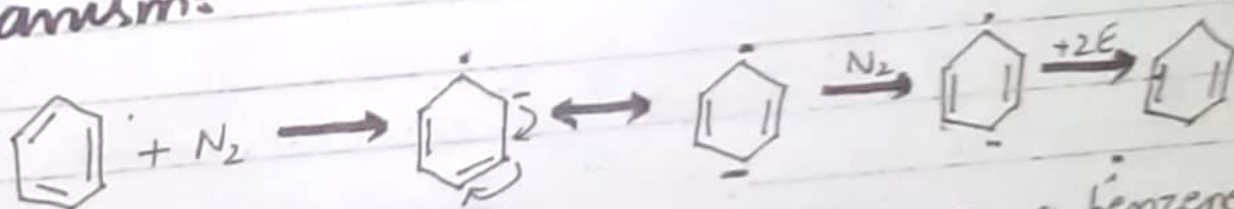


naphthalene

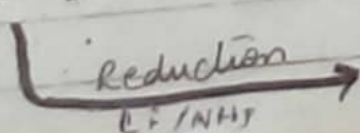
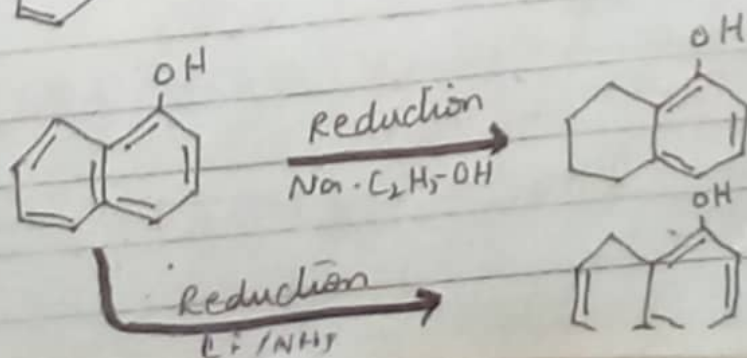
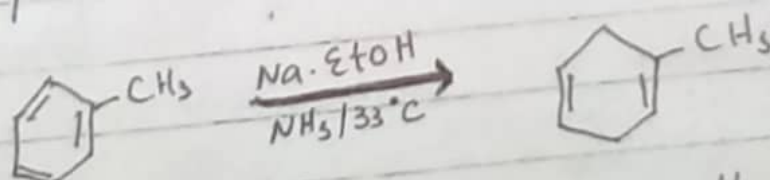


When Birch

Mechanism:



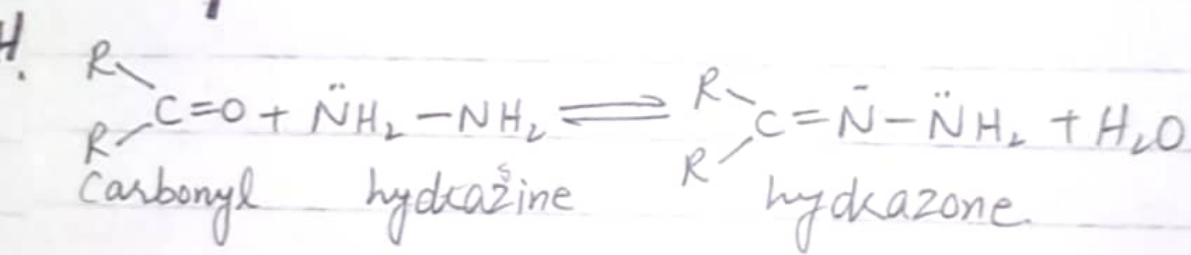
Example: When birch reduction occurs on a benzene ring with an electron withdrawing substituent. It ends up on a carbon with double bond in the product.



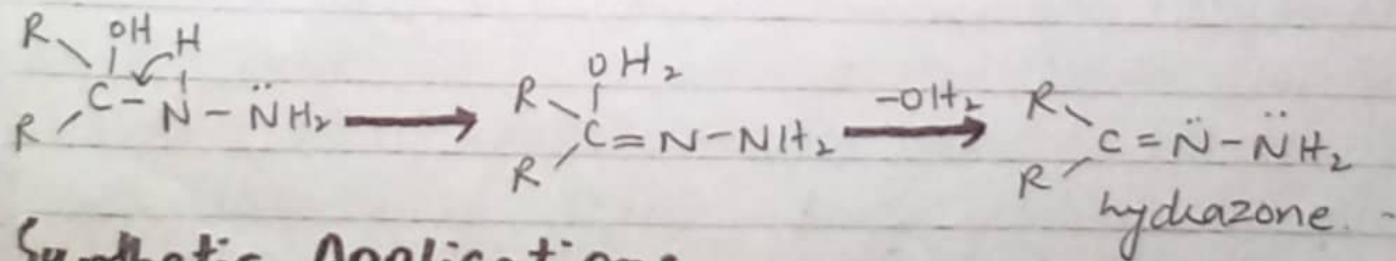
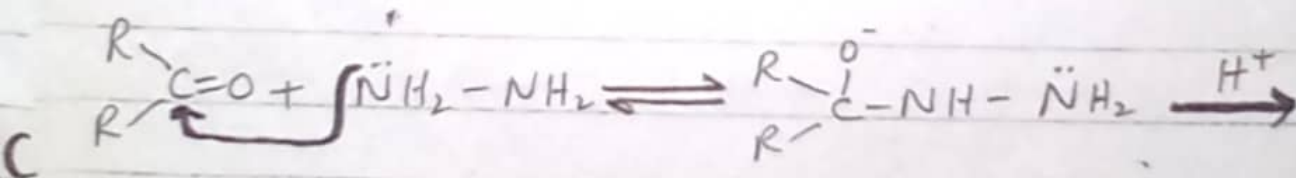
Note on Wolff-Kishner reduction with example and mechanism:

In this procedure the hydrazone is formed by heating carbonyl compounds in the presence of hydrazine, hydrazide base triethylene glycol at $180-200^{\circ}\text{C}$ for many hours. After completion of reaction water is removed from reaction mixture and resulting hydrazone is decomposed by further heating.

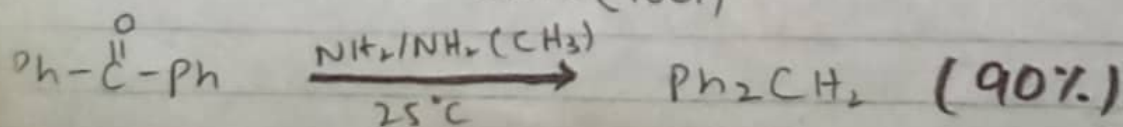
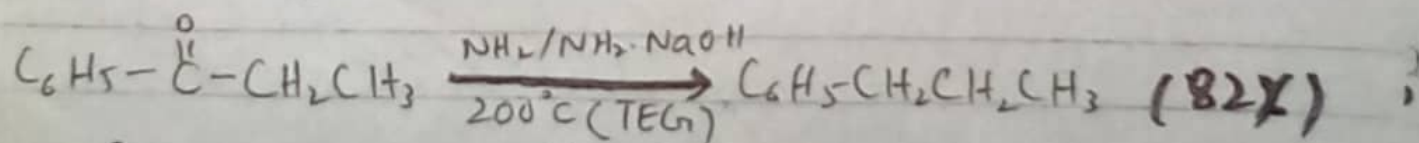
Example:



Mechanism:



Synthetic Application:

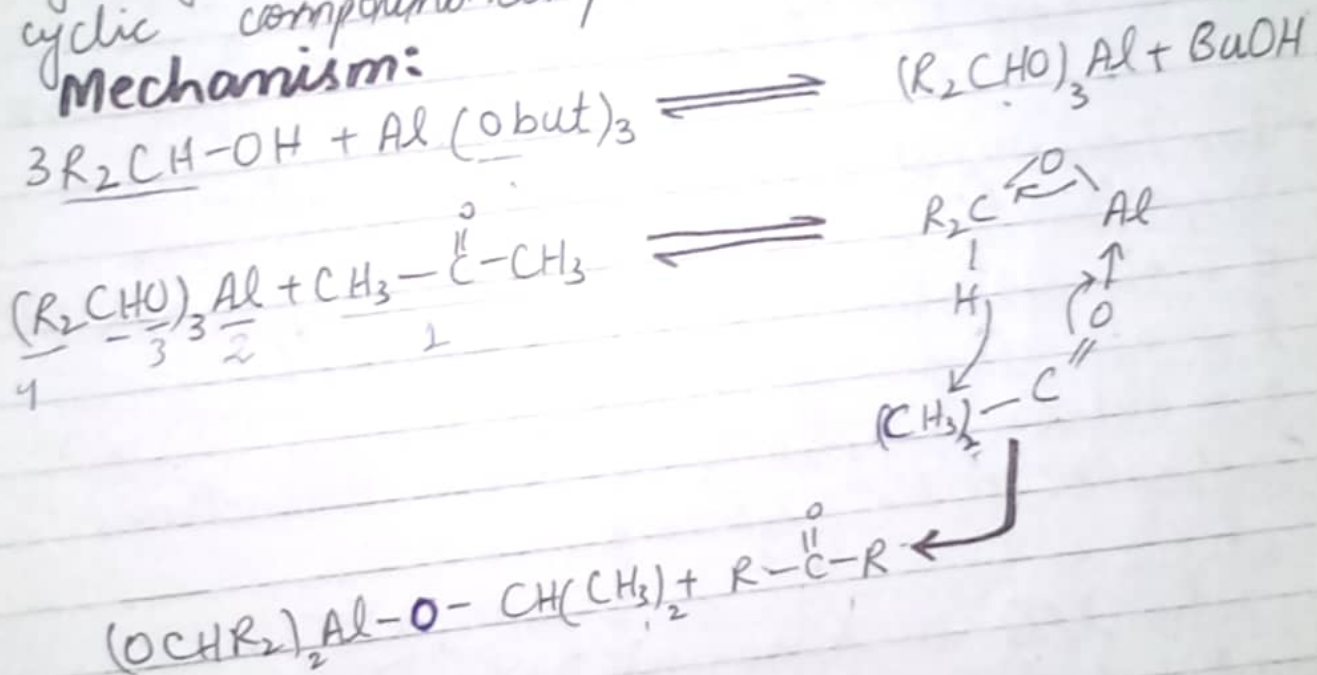


(4)

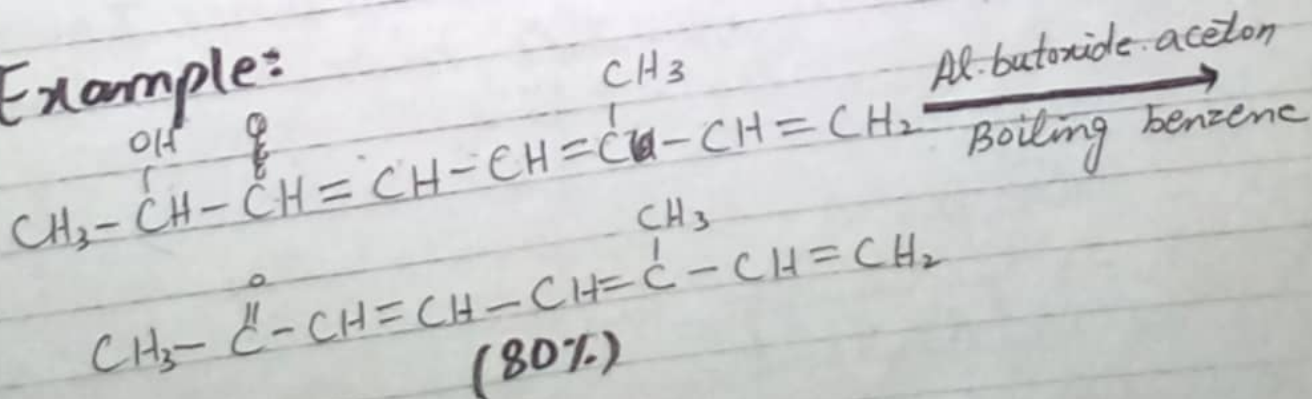
(5) Oppenauer Oxidation of secondary alcohol. Give example and mechanism:

This is very important reaction used for secondary alcohols, oxidizing agent is ketone like acetone, butanone, cyclohexanone etc. the mechanism involves deprotonation of the alcohol by equilibrium with alkoxide followed by hydride transfer to the ketone with cyclic compound complex.

Mechanism:



Example:



(2018)

① What is Beer-Lambert Law?

“Beer-Lambert law which states that the fraction of incident radiation absorbed is proportional to the number of absorbing molecules in its path.”

When the radiation passes through a solution the amount of light absorbed or transmittance is an exponential function of the molecular concentration of the solute and also a function of length of the path of radiation through the sample. Therefore

$$\log \frac{I_0}{I} = \epsilon c l$$

I_0 = Intensity of incident light

I = Intensity of light transmitted.

c = Concentration of solute in mol/L

l = Path length of sample in cm

ϵ = molar absorptivity or molar extinction coefficient of the substance.

The ratio I/I_0 is known as transmittance T and logarithm of inverse ratio I_0/I known as absorbance A .

$$-\log \frac{I}{I_0} = -\log T = \epsilon c l$$

$$\log \frac{I_0}{I} = A = \epsilon c l$$

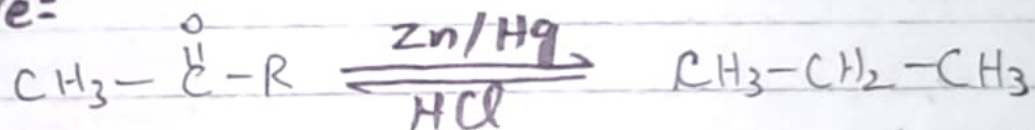
$$\boxed{A = \epsilon c l}$$

(3) Clemmensen reduction with eg and mechanism

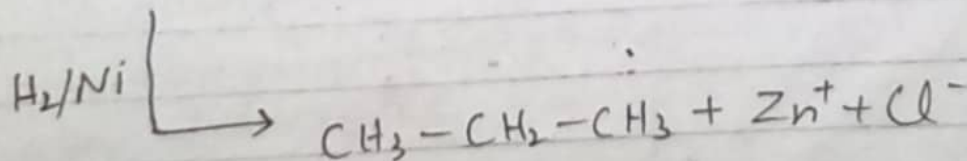
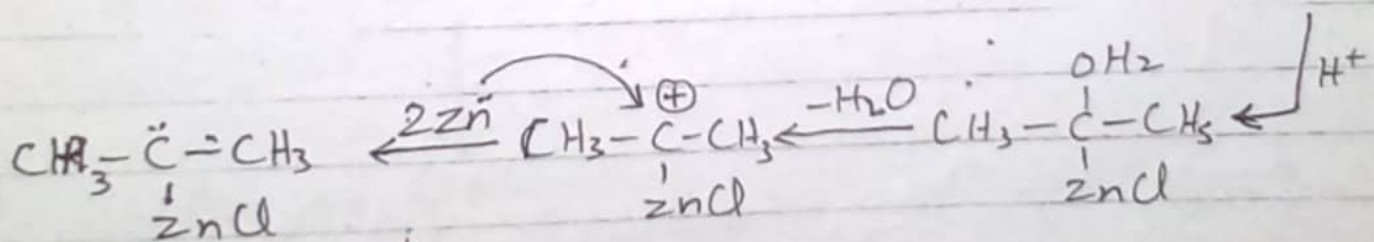
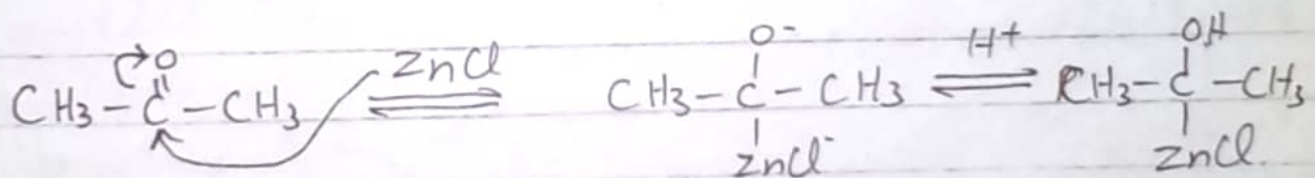
"The carbonyl group of aldehyde and ketone is change into methyl or methylene. When react with zinc amalgam in the presence of HCl. This method is known as clemmensen reduction."

→ The reaction mixture consist of carbonyl compound, amalgamated zinc and concentrated HCl. (It is a source of Hydrogen)

Example:



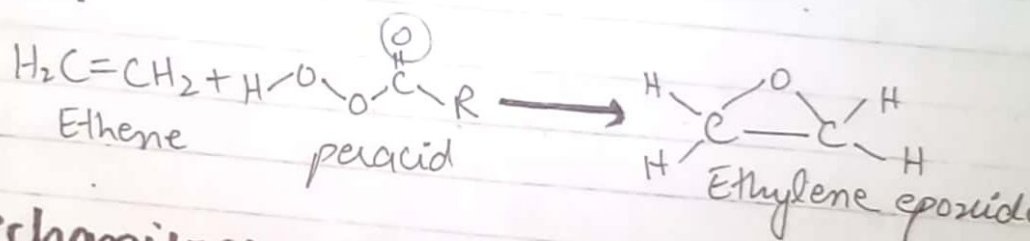
Mechanism:



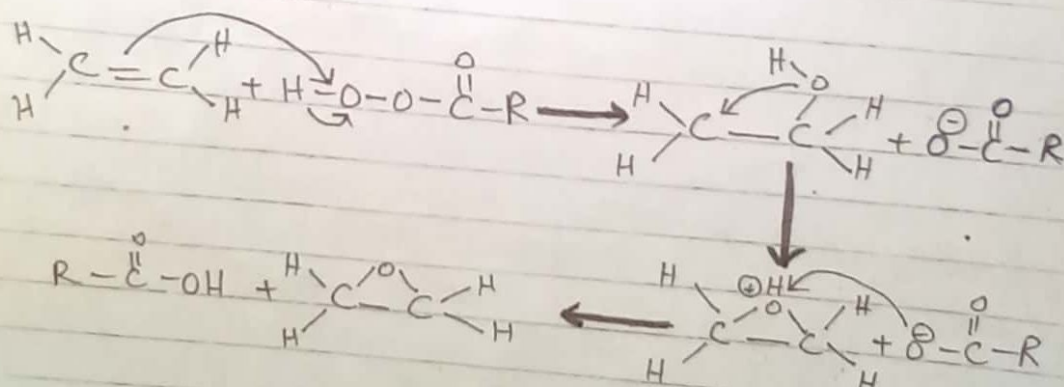
(4) Epoxidation of alkenes with e.g and mechanism:

“The type of oxidation in which an oxygen atom is transferred from the peracid to the C=C double bond thus forming an oxirane ring called epoxidation”
→ The transfer oxygen atom carries +ve charge peracid must be considered electrophilic oxidation. The alkaline H_2O_2 used for epoxidation.

Example



Mechanism:



(5) What is effect of Ring size on IR absorption in cyclic ketones:

The size of ring affects the $C=O$ stretching frequency. The frequency increases as the ring gets smaller. This is because small rings have more p-character in their ring bonds. Which means the bonds outside the ring have more s-character and are stronger.

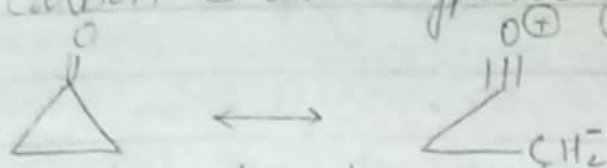
The carbonyl carbon prefers to be (more or less) sp^2 hybridized. If two bonds attached to the carbonyl carbon and bending them toward each other so that the $R-C-R$ angle is forced from $120-90^\circ$ so the hybridization of the orbital the carbonyl carbon uses to form these bonds changes from sp^2 to p . As these bonds use more p-character more s-character remains for use in the $C-O$ sigma bond. This would strengthen the CO bond and raise the energy required to stretch it (increasing its IR stretching frequency).

Resonance also play an important role on IR stretching frequency

For Example:

When we strain the carbonyl by placing it in a small ring the bonds b/w the carbonyl carbon and the adjacent carbon becomes weaker. This means that resonance structure with a strong triple bond ($C \equiv O$) becomes more important which

Involves Carbon-Carbon hyperconjugation



and on the other hand cyclohexanone in its resonating structure leads to a weakening of the carbonyl π -bond that's why conjugating decreases $C=O$ ring strain in IR absorption.

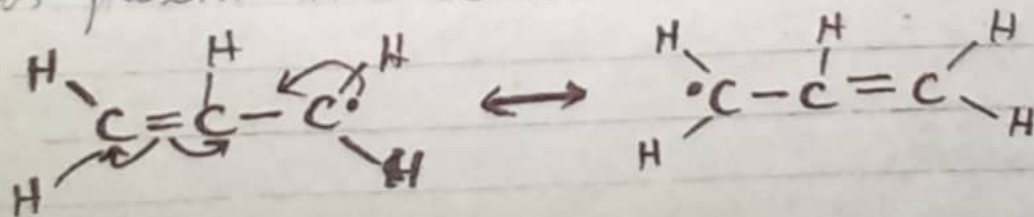


(2) Factors affecting Stability of Free Radical:

Free radicals are those species which are highly reactive and are unstable. But there are some factors which increase the stability of Free Radicals like:-

1. Resonance:

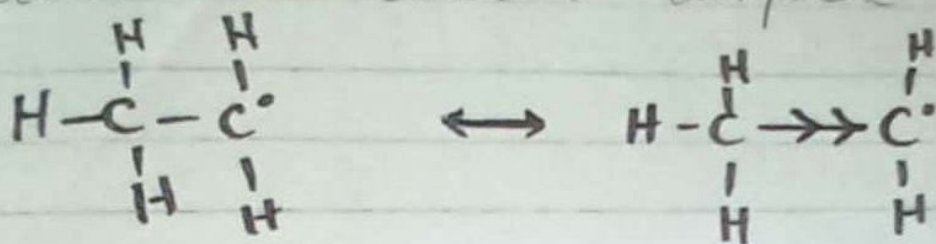
Resonance is a process or phenomenon where the molecule is more stable by the delocalization of π -electron. More Resonance is present in Radical more stable the radical is.



2 Inductive effect:

Free Radical consist of unpaired

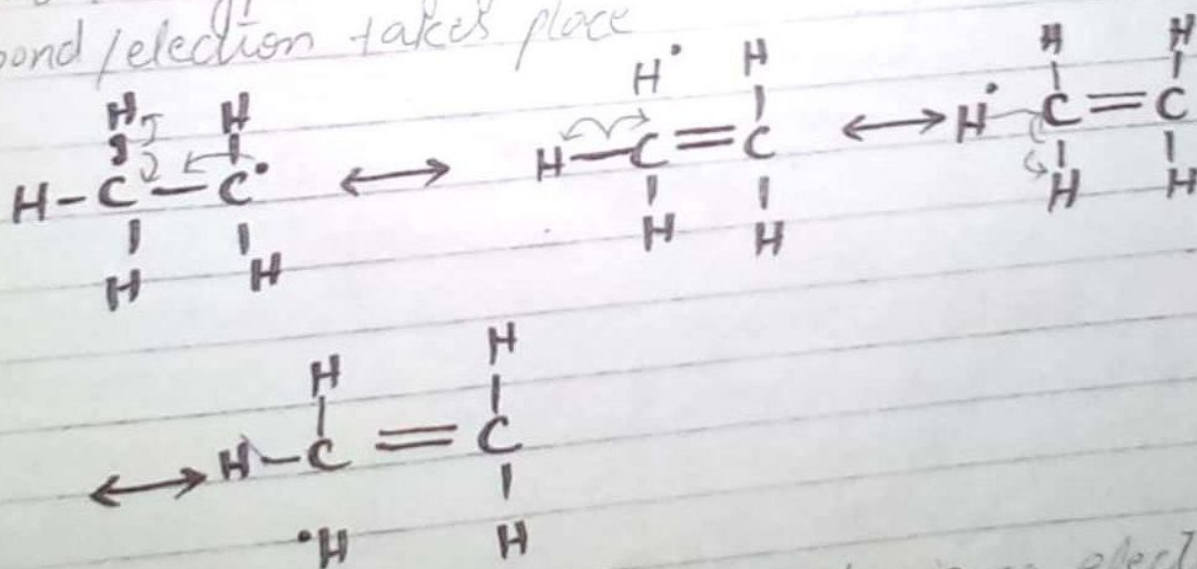
electron it doesn't complete its octet



Carbon forming an radical and attached with a methyl group which show **+I effect** it will try to push the electron to the carbon which have are deficient of electron making it stable

3- Hyperconjugation:

This phenomenon is very similar to Resonance the only difference is in resonance the delocalization of π -electron/bond takes place and in hyperconjugation the delocalization of sigma bond/electron takes place



This shows the delocalization of sigma electron b/w carbon hydrogen more hyperconjugating structure more stability of free radicals

may be solid, liquid

Analytical Chemistry

1

9 Mar (16029) 1-

Distribution Law:

This law was first given by **Nerst** who studied the distribution of several solute b/w different appropriate pairs of solvents.

This law states as:

“A solute will distribute itself b/w two immiscible solvents so that the ratio of its concentration in each solvent is equal to the ratio of its solubility in each one.”

→ Distribution of solutes b/w immiscible solvents

$$K_d = \frac{C_o}{C_w}$$

C_o = Molar concentration in Organic layer.

C_w = Molar concentration in Aqueous layer.

K_d = Distribution constant / Coefficient.

Types of Ion Exchanger:

Ion exchanger are the resins that are composed of polymers with cross-linking.

Types:

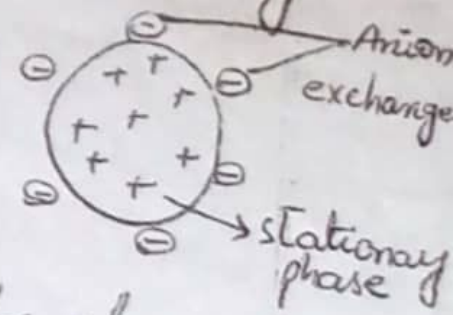
They are of two types:

1- Anion exchanger:

They are +vely charged counter labile ions available for exchange.

Example:

H^+ , Na^+ , Mg^{2+} , etc



2- Cation exchanger

They are -vely charged counter labile ions available for exchange.

Example:

OH^- , Cl^- , SO_4^{2-} , etc

